HSE INTELLIGENCE PLATFORM · FOR CLIENT REVIEW

How Artificial Intelligence Works in Your Platform

Where AI appears in the web console and the mobile apps, how every answer is grounded in your own live data, the safeguards that prevent it inventing figures, and how it aligns to your AI-ISMS specification.

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Aligned to *AI-ISMS Data Architecture & AI Engine Ver 1.0*

ADVISORY ONLY | FULLY AUDITABLE

In plain terms

The AI in your platform is an **advisor, not a decision-maker**. It reads your organisation's live safety data, answers questions about it, and suggests actions. A person always decides.

The most important design choice is this: **every time anyone asks the AI a question, the system first pulls a fresh snapshot of your real data** — incidents, near misses, permits, CAPA status, compliance ratings, contractor risk and your safety KPIs — and hands it to the AI along with the question. The AI is then instructed, in writing, that it may only use figures from that snapshot.

This means two things that matter to you. First, **the AI cannot invent a number**; if something isn't tracked, it must say so rather than estimate. Second, **its answers always match what you see on your dashboards**, because both are reading the same data at the same moment.

All of the safeguards run on the server. Neither the web app nor the mobile app can switch them off or change the AI's instructions.

1
Single secure AI service shared by web and mobile — one set of rules, applied identically

0
Figures the AI is permitted to invent. It must cite your data or state that it isn't tracked

8
Engine functions in your specification, mapped to the platform below

2
AI providers configured, with automatic failover — the platform keeps working regardless

Built for the full scope in your specification
Predictive injury risk scoring · control-of-work enforcement · ISO 45001 compliance and auditability · contractor and high-risk work control · multimodal input across text, image, video and audio · explainable, regulator-safe output.

WEB CONSOLE — Where you meet the AI

For Safety Managers, Directors and Administrators — analysis, oversight and reporting

AI Safety Advisor — DELIVERED
Full conversational assistant. Ask anything about your safety position and receive an answer grounded in current figures.

Violation Analysis — DELIVERED
Identifies trends, high-risk areas and the top recommended actions from your violation record.

Risk Forecast — DELIVERED
Forward view of likely risk with preventive actions, drawn from your historical data.

Zone Risk Assessment — DELIVERED
Zone-by-zone risk picture with controls recommended in priority order.

Recommendations — DELIVERED
Prioritised HSE improvement recommendations for any selected context.

Model Governance View — NEXT PHASE
Visibility of which AI model version is in use, what it was trained on, and the ability to roll back.

MOBILE APPS — Where you meet the AI

For Workers, Supervisors, Safety Managers and Auditors — guidance at the point of work

Worker · Safety Assistant — IN PROGRESS
Ask safety questions on site — PPE requirements, procedures, hazards. **Interface is complete and the AI service behind it is live; connecting the two is scheduled for the next release.**

Supervisor · Safety Insights — IN PROGRESS
Already shows live injury-risk score, safety KPIs and compliance alerts from real data. AI-generated interpretation of those figures is the next step.

Manager · Action Suggestions — NEXT PHASE
When assigning a corrective action, the AI proposes suitable engineering, procedural, supervisory or training controls in priority order.

Auditor · Checklist Assist — NEXT PHASE
Highlights the checks that most often reveal real problems, and flags items that add little value.

Why mobile follows the web console The AI engine, the data grounding and every safeguard are already built and running — they were proven on the web console first. Bringing them to mobile is now an interface exercise rather than new engineering, which is why these items move quickly once scheduled.

HOW A QUESTION BECOMES A TRUSTWORTHY ANSWER

The same six steps run whether the question comes from the web console or a phone on site. There is no separate mobile logic and no shortcut path. These steps implement the processing sequence set out in your specification — *ingestion, normalisation, validation, analysis, decision, record, and the learning loop.*

STEP 1 — The question is asked
A user asks a question in the web console or the mobile app. Only the conversation is sent — **the AI's instructions are never sent from the device**, so they cannot be altered.

▶

STEP 2 — Identity is verified
The user's login and organisation are confirmed. Every piece of data gathered next is restricted to that organisation — **one customer can never see another's information**.

▶

STEP 3 · THE CRITICAL STEP — Your live data is attached
A fresh snapshot of your real figures is built **on every single question** — incidents, near misses, permits, CAPA, compliance ratings, contractor risk and safety KPIs. Nothing is cached, so nothing is out of date.

▶

STEP 4 — Safeguards are applied
The AI receives its written rules alongside your data — cite real figures only, never invent, label predictions as estimates, stay within safety scope.

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STEP 5 — The answer is produced
The request goes to the primary AI provider, with automatic failover to a second. If neither is available the platform continues to run normally.

▶

STEP 6 · NEXT PHASE — The answer is recorded
Each answer stored with a confidence score and marked as AI-generated, then the user's decision to accept, amend or reject it is captured — creating a full audit trail.

STEP 3 IN DETAIL · WHAT THE AI IS ALLOWED TO READ

Organised exactly as your AI-ISMS specification defines it, so you can see which of the four data classes are feeding the AI today and which are scheduled.

CLASS A NEXT PHASE
Standards & Reference Knowledge
Supplied with the platform
- ISO 45001, ISO 14001, COSHH, OSHA, IOGP and related codes
- Investigation methods — 5-Why, TapRooT, Tripod Beta
- Will allow the AI to **cite a specific clause** rather than reason from numbers alone

CLASS B IN PROGRESS
Your Policies & Libraries
Maintained by you in the web console
- Organisation structure, sites, roles — **available now**
- Hazard and risk libraries — **available now**
- Policies, SOPs and work instructions — held in the platform; being prepared so the AI can quote them directly
- Competence profiles — arrive with the Training workflow

CLASS C LIVE TODAY
Your Live Operations
Produced by your teams in the field
- Incidents, near misses, permits, safety walks, CAPA
- Safety KPIs — **TRIR, LTIFR, DART, FAR, near-miss ratio**
- Compliance, risk and contractor risk position
- **Rebuilt on every question — never cached, never stale**

CLASS D NEXT PHASE
Learning & Improvement
Makes the system better over time
- Anonymous industry benchmarking
- Learning from where your team accepts or rejects AI advice
- Model version control and rollback
- Checklist and root-cause pattern improvement

STEP 4 IN DETAIL · THE SAFEGUARDS

These run on the server, so both the web console and the mobile apps inherit them identically. No user, and no device, can switch them off.

Preventing invented figures ACTIVE

> *"Ground every claim in the snapshot data — cite the specific numbers you are using.* ***Never invent a figure that isn't in the snapshot.*** *If something isn't covered, say it isn't tracked yet rather than guessing."*

- This is the actual instruction given to the AI on every request
- It forces the AI to **admit a gap** instead of producing a plausible-sounding number
- Directly implements the anti-hallucination requirement in your specification

Keeping answers honest and useful ACTIVE
- Any forecast must be **clearly labelled an estimate**, never stated as fact
- Answers must focus on leading indicators and preventive action
- Answers must be specific — vague disclaimers are explicitly disallowed
- Concise by default, with detail on request

Your specification's limits, enforced in software NEXT PHASE
- Does **not** make autonomous safety decisions
- Does **not** discipline individuals
- Does **not** replace human accountability
- Does **not** use medical or biometric data without explicit consent
- **Never overrides a safety gate.** Where the system hard-blocks work — an expired safety-critical certificate, or a driver over the fatigue limit — the AI may advise, but it cannot lift that block

ALIGNMENT TO YOUR AI-ISMS SPECIFICATION · THE EIGHT ENGINE FUNCTIONS

Each of the eight functions defined in your specification, with where it appears in the platform and its current status.

FUNCTION 1 — IN PROGRESS
Policy, KPI & SOP Compliance
Checks your safety intent and controls against standards and best practice, reporting gaps and weak controls.
Web console · completes when policies are made citable

FUNCTION 2 — IN PROGRESS
Risk Identification & Scoring
A living risk picture and heat map that updates as conditions change and controls weaken.
Web risk register · mobile hazard register

FUNCTION 3 — DELIVERED
Leading KPI Generation
Moves measurement from lagging to leading indicators — TRIR, LTIFR, DART, FAR and near-miss ratio are calculated and fed to the AI today.
Web analytics · mobile insights

FUNCTION 4 — IN PROGRESS
Predictive Injury Risk Scoring
An injury-risk score is live now. Your specification's fuller model — five domains with clear driver attribution over 7, 30 and 90 days — is the next step.
Both clients

FUNCTION 5 — NEXT PHASE
Corrective Action Suggestion
Proposes engineering, administrative, procedural, supervisory or training controls, prioritised by severity and feasibility.
Mobile, at the point of assigning actions

FUNCTION 6 — NEXT PHASE
Control-of-Work Oversight
Watches permit compliance, bypasses and simultaneous operations, escalating when work drifts from plan. Camera-based monitoring remains phase two.
Mobile permit workflow

FUNCTION 7 — NEXT PHASE
Communication & Leadership
Reviews toolbox talks and briefings for clarity, follow-through on commitments and visible leadership engagement.
Mobile · requires toolbox talk capture

FUNCTION 8 — NEXT PHASE
Continuous Learning
Improves from control effectiveness, your team's decisions on AI advice, and anonymous benchmarks. Versioned and auditable throughout.
Platform-wide

↺ HOW THE SYSTEM GETS BETTER OVER TIME

Your specification identifies human feedback as the core feature of the learning design. This is how it works in practice.

THE CORE FEATURE — Your team teaches it
Whenever someone accepts, amends or rejects an AI suggestion — or overrides a safety gate — the reason and the outcome are recorded. Over time this reduces false alarms and builds frontline trust.

ACCURACY REVIEW — Predicted against actual
After a significant event, or one narrowly avoided, the predicted risk is compared with what actually happened and the model's weighting is adjusted.

CHECKLIST IMPROVEMENT — Sharper inspections
Checks that consistently reveal real problems are promoted; those that rarely find anything are removed. Your audit results drive this directly.

GOVERNANCE — Controlled and reversible
Every model version is recorded with its change history. Updates are never silent, and any previous version can be restored.

DELIVERY STATUS

An honest position on what is working now and what follows.

CAPABILITY	STATUS	WHERE IT APPEARS	WHAT IT MEANS FOR YOU
WORKING TODAY			
AI Safety Advisor with conversation history	DELIVERED	Web console	Ask follow-up questions naturally; the assistant retains the thread
Live data attached to every question	DELIVERED	Platform-wide	Answers always match your dashboards, because both read the same data at the same moment
Anti-invention safeguards	DELIVERED	Platform-wide	The AI must cite your real figures or state that something isn't tracked
Two AI providers with automatic failover	DELIVERED	Platform-wide	No single point of failure; the platform runs normally even with no AI available
Violation, risk, zone and recommendation analysis	DELIVERED	Web console	Four ready-made analyses, no prompting skill required
Safety KPIs feeding the AI	DELIVERED	Web & mobile	TRIR, LTIFR, DART, FAR and near-miss ratio are computed and available to the AI
SCHEDULED NEXT — IN PRIORITY ORDER			
Mobile Safety Assistant connected to the AI service	IN PROGRESS	Mobile · Worker	Puts the same grounded assistant in the hands of the people doing the work. Interface and engine both exist
AI answer log with confidence scoring	NEXT PHASE	Platform-wide	Every AI answer becomes auditable — required for regulatory defensibility
Capturing accept / amend / reject decisions	NEXT PHASE	Web & mobile	The learning loop your specification identifies as its core feature
Five-domain predictive injury risk scoring	IN PROGRESS	Platform-wide	Not just a score, but a clear explanation of *why* risk is rising and where to intervene
Standards and policy library made citable	NEXT PHASE	Platform-wide	The AI can quote the specific ISO clause or your own SOP, not just interpret numbers
Model governance and version control	NEXT PHASE	Web console	Full visibility and rollback — needed before any regulated deployment

PREDICTIVE INJURY RISK SCORING · THE FIVE LEADING-IMPACT DOMAINS

Your specification defines injury risk as the product of five measurable domains, each built from leading indicators rather than injury counts — so risk is visible before harm occurs. The platform calculates these from the operational data your teams already capture.

Hazard Exposure
High-risk task frequency · exposure hours · hazard severity index · proportion of work without a valid risk assessment
How risk behaves: rises in step with exposure volume, and faster with hazard energy. A missing risk assessment applies an immediate penalty.

Control Integrity
Strength of the control hierarchy · permit condition violations · recurrence of unsafe conditions · corrective action effectiveness
How risk behaves: escalates sharply where PPE is relied on in place of engineering controls, and compounds when the same hazard returns after closure.

Work Authorisation & Discipline
Proportion of work under a valid permit · permit bypass rate · quality of permit closure · toolbox talk completion
How risk behaves: spikes with unpermitted work, late or poor closures, and repeated discipline breaches.

Human Readiness & Capacity
Task-competence mismatch · training and induction gaps · fatigue index · exposure of new or inexperienced workers
How risk behaves: multiplies hazard risk rather than adding to it. Fatigue combined with a competence gap produces a step change.

Organisational & System Health
Near-miss to injury ratio · corrective action ageing · supervisor safety engagement · recurring human-factor failures
How risk behaves: accumulates over time. Ageing actions and repeating system failures raise the probability of injury clusters.

Two scores, one model. The same five domains support both approaches in your specification. The **deterministic score** applies fixed published weights to measure where safety stands right now — every figure traceable to an explicit rule an auditor can inspect. The **predictive score** uses the same domains to estimate the probability of a recordable or lost-time injury over 7, 30 and 90-day horizons, and returns **clear driver attribution** explaining why risk is rising. The deterministic score is the auditable core; the predictive score is the intelligence layer above it.

Delivered · In progress · Next phase

The engine recommends; a person decides. Every output is advisory, traceable to your own data, and requires human validation — and the AI can never lift a safety block raised by the competence, fatigue, contractor or weather checks.